

MotionPI System Visualization — Milestone Process Book

Course: Visualization for Data Science • Project: final-project-kinetictrace

Overview & Motivation

This project builds an interactive web visualization to integrate and inspect multiple data streams from a mobile-health study: ENMO activity (accelerometer), PA/EMA survey triggers, app/system events, and device reliability signals (battery, storage). The goal is to verify activity-based PA/EMA triggers, diagnose reliability issues (e.g., data gaps from battery or disconnects), and align participant events with activity patterns for rapid review by HCI and CS collaborators.

Audience

Primary users: ~20 research collaborators across medicine, public health, geography, and CS. Design emphasizes clarity for non-CS users handling complex, multi-stream data.

Data

Stream	Notes
ENMO activity	1–2s cadence (auto-detected ~2s in sample); 1 Hz nominal; wristband accelerometer.
Events / logs	Survey triggers, data collection toggles, connect/disconnect, low battery, etc.
Battery & storage	Phone battery %, left/right wrist battery %, wrist storage used %.
Timestamps	Localized to America/Denver for display.

Exploratory Data Checks

Cadence detection via median timestamp delta: ~2 seconds. We compute a 7-minute moving average (centered) over ENMO and check a PA/EMA rule: $ENMO \geq 0.1006$ for $\geq 70\%$ of samples within a 7-minute window. On this sample, the fraction of windows meeting the rule was ~0%, which is consistent with low activity segments; the visualization still renders all layers for inspection.

Visualization Design

Layers: (1) ENMO scatter (WebGL) + 7-min moving average (red) + threshold (gray); (2) PA/EMA visuals (green pre-window band + trigger line/arrow); (3) Events with stable color/symbol mapping and 30s deduplication; (4) Battery & storage on right y-axis; (5) Day/Night shading with ☀/🌙 icons. A compact floating control panel supports pan/zoom, quick range presets, PNG export, and per-layer toggles. Defaults show ENMO only to reduce clutter (focus+context). Color-blind-friendly palettes and redundant symbols improve accessibility.

Implementation Highlights

Implementation uses Python (pandas, Plotly) to generate a standalone HTML (src/index.html). WebGL scatter ensures responsiveness on 100k+ points. Data is resampled to cadence; event markers are placed on an overlay axis to remain visible, and wrist battery/storage are auto-matched to left/right by MAC. The control panel is injected via a postscript for pan/zoom, Y-scale presets, and layer toggles.

Screenshots (Milestone)

Add screenshot of src/index.html (overview)

Add screenshot zoomed to a PA■EMA window

Evaluation (Milestone)

Task success: The viz allows verifying PA■EMA detection and relating failures to reliability overlays.

Usability: common questions can be answered with ≤ 3 interactions (toggle layers, zoom, read markers).

Performance: WebGL scatter + downsampling provides smooth pan/zoom on large traces.

What Works Now

- All core layers render (ENMO, MA7, threshold, PA■EMA bands/lines, events, battery/storage, day/night).
- Event de■duplication (30s) and stable color/symbol mapping.
- Wrist side auto■match by MAC; overlay axis keeps event markers readable.
- Downsampling and WebGL ensure responsiveness.

Next Steps

- Add drag■and■drop CSV loader; multi■participant dropdown.
- Anomaly flags for long gaps, repeated disconnects, or low storage.
- Refine legends, add help tooltips, and export presets.
- Broader evaluation with collaborators; document edge■case parsing.

Milestone Checklist

- Prototype HTML at src/index.html (working locally).
- Process book (this PDF) added to repo root.
- README updated with run/view instructions and data file list.
- GitHub release tagged (e.g., milestone-v1).